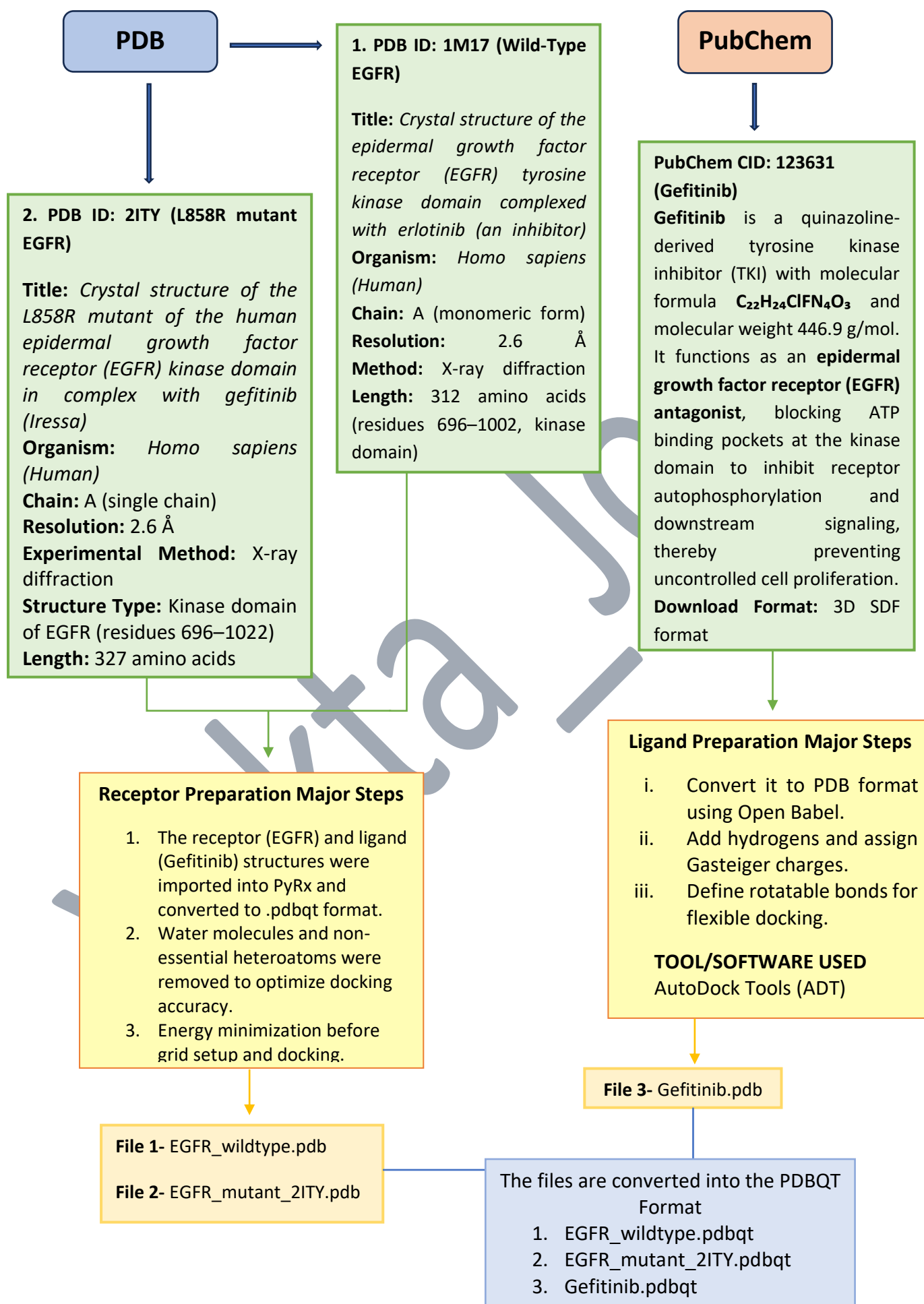


2.2. WORK FLOW OF THE PROJECT



Molecular Docking

- i. Load receptor (wild/mutant EGFR) and ligand (Gefitinib) PDBQT files into PyRx.
- ii. Visualize both molecules in the 3D viewer.
- iii. Set the grid box around the ATP-binding pocket (using coordinates from the reference complex).
- iv. Run AutoDock Vina to perform docking.
- v. Record the binding affinities and RMSD values of the top poses.

Software Used-
(PyRx)

Molecular Visualization (PyMOL)

- i. The best docked complex from PyRx is imported into PyMOL for visualization.
- ii. Receptor (wild or mutant EGFR) and ligand (Gefitinib) are loaded to view the binding pose.
- iii. The complex is visualized in cartoon and surface formats to observe ligand fitting.
- iv. Key interacting residues are identified to confirm stable and accurate binding at the ATP site.

Software Used-
(PyMOL)